

Ethical AI Evaluation Report

Project: StressLens **Generated:** July 30, 2026

 **DATA QUALITY NOTICE:** The following data integrity issues were detected. Scores may be suppressed.

Cumulative Risk Volume

16.50

Based on 37 quantitative questions
(Maximum possible cumulative volume: 148.00)

Overall Average ERC (Mean): 0.27 / 4

Max Principle Average (Sensitivity): 0.44 / 4

 Overall risk promoted due to Max Principle Sensitivity override.

(ERC values are normalized on a 0–4 scale)

Final Risk Label:

Minimal Risk

Ethical Tensions

2 IDENTIFIED

2 Accepted, 0 Under Review

Total Questions Assessed: 37 total (37 Quantitative, 0 Qualitative)

0 qualitative (open-text) questions are excluded from quantitative risk scoring. These questions provide narrative insights that complement the quantitative analysis.

*Note: 0 qualitative (open-text) questions are excluded from quantitative risk scoring.
Quantitative risk assessment is based on 37 questions with predefined answer options.*

Ethical Principles Risk Overview

Ethical Principle	Cumulative Risk Volume	Question Count	Average ERC (/ 4)	Risk Level
TRANSPARENCY	4.00	9	0.44 / 4	Minimal Risk
HUMAN AGENCY & OVERSIGHT	1.50	8	0.19 / 4	Minimal Risk
TECHNICAL ROBUSTNESS & SAFETY	2.00	13	0.15 / 4	Minimal Risk
PRIVACY & DATA GOVERNANCE	2.00	8	0.25 / 4	Minimal Risk
DIVERSITY, NON-DISCRIMINATION & FAIRNESS	0.00	3	0.00 / 4	Minimal Risk
SOCIETAL & INTERPERSONAL WELL-BEING	1.50	6	0.25 / 4	Minimal Risk
ACCOUNTABILITY	5.50	14	0.39 / 4	Minimal Risk

Qualitative Analysis of Open-Text Responses

Methodology

In addition to quantitative scoring, expert evaluators provided open-text responses covering technical, legal, and ethical dimensions of the StressLens system.

The following analytical summaries synthesize expert observations with quantitative risk assessments, highlighting structural vulnerabilities and ethical tensions across all core principles.

TRANSPARENCY

In response to Q9 regarding explainability, experts observed that while the HUD displays probability values indicating ML generation, 'model limitations, false alarm rates, and accuracy levels are not systematically communicated to the user.' This reveals a structural vulnerability where raw probabilistic scores are presented without essential contextual framing, increasing the risk of misinterpretation. This opacity directly contrasts with the quantitative risk score by obscuring algorithmic blind spots from end-users and operators.

HUMAN AGENCY & OVERSIGHT

Addressing Q10 regarding human control, experts stressed that 'no mechanism is defined allowing users to challenge or override AI decisions.' Stress warnings are generated automatically without a procedure for users to reject predictions or disable the system, while operators lack real-time intervention capabilities. This structural imbalance strips individuals of agency and fosters automation bias, amplifying the risks associated with unverified biometric classifications.

TECHNICAL ROBUSTNESS & SAFETY

Evaluating technical resilience, experts noted that the system fails under specific conditions such as 'contact lenses, makeup, dry skin, sensor disconnection, or calibration failure.' Furthermore, WebSocket communication relies on unencrypted ws:// protocols. These vulnerabilities reveal inadequate defensive engineering and fallback mechanisms, directly undermining system reliability in active VR environments.

PRIVACY & DATA GOVERNANCE

Legal experts highlighted in Q4 that 'StressLens processes eye-tracking and EDA data, which constitute special category personal data under GDPR Article 9 and KVKK Article 6,' yet lack explicit legal bases, privacy notices, or DPIA documentation. Storing raw data in unencrypted CSV files without retention limits violates data minimization principles and exposes organizations to severe regulatory liability.

DIVERSITY, NON-DISCRIMINATION & FAIRNESS

Experts observed that mitigation measures are limited to SMOTE and session calibration, noting that 'fairness metric analysis, independent bias audits, and disparate impact evaluations have not been performed.' The unknown demographic diversity of the training set creates a high structural risk of misclassification and discriminatory impacts against specific user groups.

SOCIETAL & INTERPERSONAL WELL-BEING

In evaluating broader societal impacts, experts warned that utilizing stress data in performance evaluations 'can undermine employee-employer trust and lead to professional stigmatization.' Continuous physiological monitoring generates surveillance pressures that erode psychological safety within organizational settings.

ACCOUNTABILITY

Governance evaluations revealed that 'accountability and responsibility are neither defined nor implemented.' The absence of an AI Ethics Board, formal complaint channels, and clear liability distribution among developers, operators, and employers represents a critical structural deficiency prior to deployment.

Disclaimer: The qualitative insights presented above are intended to provide contextual understanding and identify governance gaps in alignment with ISO 42001 requirements.

Executive Summary

- The assessment identified a Cumulative Risk Volume of 33 across 37 evaluated quantitative questions.
- The Normalized Average ERC is N/A / 4. According to the EU AI Act classification framework, this represents a High Risk system due to the processing of biometric data and emotion recognition systems in the workplace.
- - ISO 42001 System Context & Key risk driver: The system processes special category biometric data (eye-tracking and galvanic skin response) in an occupational setting without an established AI Management System or comprehensive DPIA.
- - Key strength & NIST Measure: The system employs a structured data processing pipeline and attempts session-level personal calibration to account for individual baseline differences.
- - Context & Governance statement: To achieve ISO 42001 certification and comply with the EU AI Act, the organization must implement rigorous data protection policies, external bias audits, and comprehensive human oversight mechanisms.

Ethical Tensions

Conflict	Severity	State	Consensus	Evidence
Safety <-> Human Oversight	high	Accepted	100%	1 items
Privacy <-> Accountability	high	Accepted	100%	1 items

Methodology & Evidence Coverage

This report matches the Ethical AI Analysis methodology for ethical AI evaluation.

Risk Calculation

- **Question Risk (ERC):** Importance (0-4) x Unmitigated Ethical Risk (0-1).
- **Cumulative Risk Volume:** Sum of all ERC contributions. Used to understand total magnitude of risk for the project.
- **Normalized Ethical Risk Level:** Determined by the higher of the Overall average ERC and the Maximum individual Principle average. This promotes the risk label if any single principle (e.g. Accountability) identifies a critical failure, even if the overall project volume is low.

Note: All numeric metrics are deterministic and traceable to MongoDB data.

Appendix: Evaluators

The ethical risk metrics aggregate all expert responses at the question level. While individual experts may differ, the normalized score reflects the combined judgment across evaluators rather than any single opinion.

Name	Role	Status
Ethical Expert	ethical-expert	Submitted
Technical Expert	technical-expert	Submitted
Legal Expert	legal-expert	Submitted

Improvement Recommendations

Short-Term (0-3 Months)

- Migrate all WebSocket communications from ws:// to wss:// (TLS) [ISO 42001: Annex A]
- Establish basic data minimization protocols and automated deletion schedules for CSV logs [NIST RMF: Manage]
- Implement user-facing disclaimers regarding model limitations and false positive rates [ISO 42001: Clause 8.2]

Medium-Term (3-12 Months)

- Integrate runtime XAI methods (such as SHAP or LIME) to explain individual predictions to operators [NIST RMF: Measure]
- Perform disparate impact analyses across diverse demographic groups to identify and mitigate bias [NIST RMF: Map]
- Conduct a formal Data Protection Impact Assessment (DPIA) and Fundamental Rights Impact Assessment (FRIA) [ISO 42001: Clause 8.1]

Long-Term (12+ Months)

- Establish an internal AI Ethics Board with authority over deployment, rollback, and incident management [ISO 42001: Clause 5]
- Deploy continuous model drift monitoring and automated performance degradation tracking infrastructure [ISO 42001: Clause 9.1]
- Obtain ISO 42001 AI Management System certification following independent third-party auditing [ISO 42001: Clause 10]

Conclusion

The evaluation of StressLens revealed a Cumulative Risk Volume of 33, classifying the system as High Risk under the EU AI Act framework due to its deployment of biometric categorization and emotion recognition technologies in occupational settings. While the system demonstrates technical initiative through session calibration, quantitative and qualitative findings emphasize profound governance and data protection gaps.

Key strengths include structured physiological data collection pipelines, whereas critical vulnerabilities center around unencrypted data transmission, absence of user recourse mechanisms, and the processing of special category personal data without a formal DPIA or FRIA. These gaps expose the organization to significant legal and ethical liabilities under GDPR and KVKK frameworks.

Moving forward, the organization must transition from an ad-hoc development model to a structured AI Management System in alignment with ISO 42001. By addressing the prioritized remediation roadmap—focusing on data encryption, human oversight, bias auditing, and rigorous governance—StressLens can successfully mitigate ethical risks and build verifiable certification readiness.

